Fundamentals Chapter 8

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1. A nice basis for V would be $\{1, x, \ldots, x^n\}$. The corresponding dual basis in V^* would be $\{\delta_{1k}, \ldots, \delta_{nk}\}$ where each δ_{ik} takes in a polynomial of degree k and returns 1 if $k = i$ and 0 otherwise. A nice basis in V^* might look like $\{f_1, \ldots, f_n\}$ where $f_i(p(x)) = a_i$, the coefficient of x^i in $p(x)$.

3a. Any element of $\bigcup \mathbb{F}[x]_{\leq n}$ is a polynomial of finite degree, so it is in $\mathbb{F}[x]$. Also, any element of $\mathbb{F}[x]$ is a polynomial of degree n for some n, so it is in $\bigcup \mathbb{F}[x]_{\leq n}$. So the two sets are equal.

3b. Let $p(x) = \sum_{i=0}^{n} a_i x^i$ and $q(x) = \sum_{i=1}^{n} b_i x^i$ where n is the maximum of the degrees of p and q. We see that

$$\phi(p(x) + q(x)) = \phi\left(\sum_{i=0}^{n} (a_i + b_i) x^i\right) = \sum_{i=0}^{n} a_i + \sum_{i=0}^{n} b_i = \phi(p(x)) + \phi(q(x)),$$

and

$$\phi(cp(x)) = \phi\left(\sum_{i=0}^{n} c a_i x^i\right) = c \sum_{i=0}^{n} a_i = c\phi(p(x)).$$

3c. Consider the function $\phi \in \mathbb{F}[x]^*$ where $\phi(p(x)) = p(1) = \sum_{i=0}^{\infty} a_i$. For contradiction, assume that $\phi \in (\mathbb{F}[x]_{\leq n})^*$ for some n. Then ϕ would only depend on the coefficients up to a_n, meaning $\phi(p(x)) = \sum_{i=0}^{n} a_i$. But ϕ sums all the coefficients, and not just up to n. So no such finite n will have the same return values as ϕ for all inputs, meaning this function cannot be equal to ϕ. So $\phi \in \mathbb{F}[x]^*$ but not $\bigcup \mathbb{F}[x]_{\leq n}$.

4. To show that $\varphi_\alpha \in V^*$ are linearly independent, we need to show that if $\sum_{q \in \mathbb{Q}} a_q \varphi_\alpha(\mathbf{v}_q) = 0$ then $a_q = 0$ for all $q \in \mathbb{Q}$. We see that the sum is only 0 if each $\varphi_\alpha(\mathbf{v}_q)$ is always 0, or if all $a_q = 0$. The former is clearly not always true, so $a_q = 0$ and φ_α are linearly independent.

7. Since a convex polygon is defined as the convex hull of finitely many points, by definition it can be expressed as a convex combination of its vertices.

8. The proof of the lemma left off by showing that if $\varphi(\mathbf{v}_i) \leq 1$ for all vertices $\mathbf{v}_i$, then

$$\varphi(s) = \varphi\left(\sum_{i=0}^{n} \lambda_i \mathbf{v}_i\right) \leq 1$$

for all $s \in S$. This is what our goal was to prove, so S^* is completely determined by the vertices of S.

9. Let's say the dual basis is $\{x, y\}$. In the xy-plane, the dual is the region satisfying the inequalities:

$$-\frac{1}{2}x - \frac{1}{2}y \leq 1,$$
$$-\frac{1}{2}x + \frac{3}{2}y \leq 1,$$
$$\frac{3}{2}x + \frac{3}{2}y \leq 1,$$
$$\frac{3}{2}x - \frac{1}{2}y \leq 1,$$

which forms a trapezoid with vertices $(0, -2), (-2, 0), (0, 2/3), (2/3, 0)$.

10. If the dual basis is $\{x, y\}$, the dual is the region in the xy-plane satisfying:

$$\pm x + y \leq 1,$$

$$\pm(2x + y) \leq 1,$$

which forms a triangle with vertices $(0, 1), (-2, 3), (-2/3, 1/3)$.

11. Consider the equilateral triangle with vertices $(2, 0), (-1, \sqrt{3}), (-1, -\sqrt{3})$. Its dual has vertices at $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), (-1, 0), \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ which lie on the original triangle.

13. The dual of the cube is the set of points that satisfy $\pm x \pm y \pm z \leq 1$, which is an octahedron. The vertices of the octahedron meet the centers of the faces of the cube.